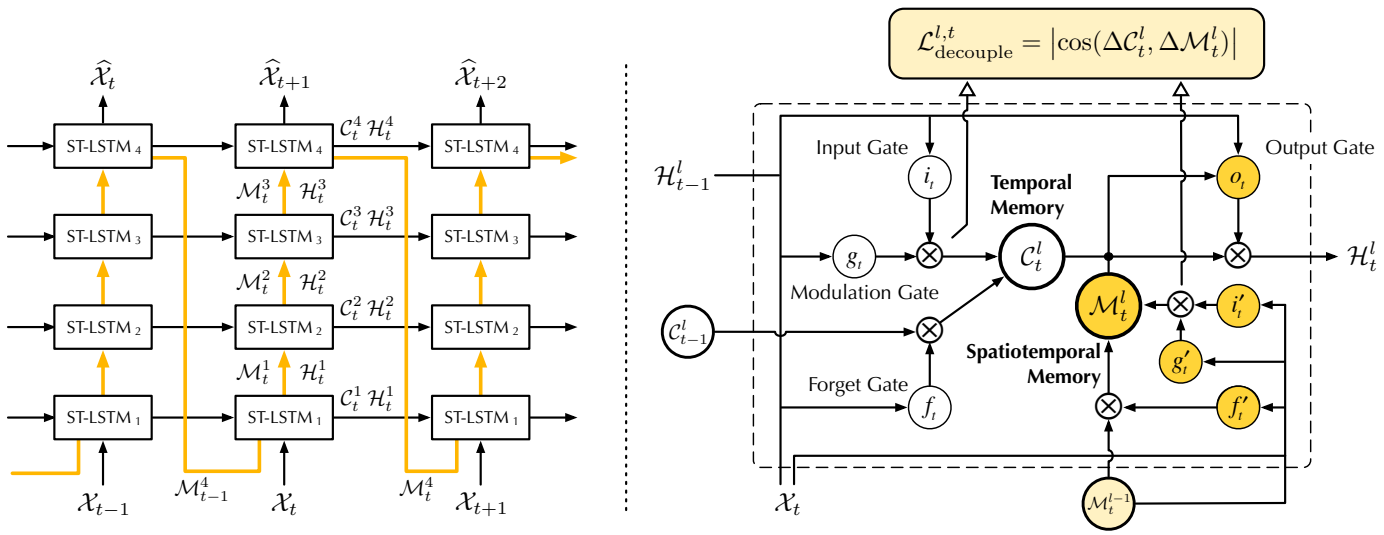

$\widehat{\mathcal{X}}_t$
$\widehat{\mathcal{X}}_{t+1}$
$\widehat{\mathcal{X}}_{t+2}$
ST-LSTM 4
ST-LSTM 3
ST-LSTM 2
ST-LSTM 1
$\mathcal{C}_t^4\ \mathcal{H}_t^4$
$\mathcal{M}_t^3$ $\mathcal{H}_t^3$
$\mathcal{C}_t^3\ \mathcal{H}_t^3$
$\mathcal{M}_t^2$ $\mathcal{H}_t^2$
$\mathcal{C}_t^2\ \mathcal{H}_t^2$
$\mathcal{M}_t^1$ $\mathcal{H}_t^1$
$\mathcal{C}_t^1\ \mathcal{H}_t^1$
$\mathcal{X}_{t-1}$
$\mathcal{M}_{t-1}^4$
$\mathcal{X}_t$
$\mathcal{M}_t^4$
$\mathcal{X}_{t+1}$
$\mathcal{L}_{\text{decouple}}^{l,t} = \left|\cos(\Delta\mathcal{C}_t^l, \Delta\mathcal{M}_t^l)\right|$
Input Gate
Output Gate
Temporal Memory
Modulation Gate
Forget Gate
Spatiotemporal Memory
$\mathcal{H}_{t-1}^l$
i_t
g_t
o_t
f_t
$\mathcal{C}_t^l$
$\mathcal{C}_{t-1}^l$
$\mathcal{M}_t^l$
i_t'
g_t'
f_t'
$\mathcal{H}_t^l$
$\mathcal{X}_t$
$\mathcal{M}_t^{l-1}$